

NAD(P)HX Dehydratase (NAXD)

Deficiency — Comprehensive Disease Characteristics Report

Disease: NAD(P)HX Dehydratase Deficiency (PEBEL2) **MONDO ID:** MONDO:0034121 · **OMIM (phenotype):** #618321 · **Gene:** *NAXD* (HGNC:25576) **Category:** Mendelian (autosomal recessive inborn error of metabolite repair)

Summary

NAD(P)HX dehydratase (NAXD) deficiency — clinically designated **PEBEL2** (Progressive Encephalopathy with Brain Edema and/or Leukoencephalopathy, type 2; OMIM #618321) — is an ultra-rare, autosomal-recessive **inborn error of metabolite repair**. It is caused by biallelic loss-of-function variants in *NAXD*, the gene encoding the ATP-dependent NAD(P)HX dehydratase (EC 4.2.1.93). Together with its partner enzyme NAD(P)HX epimerase (NAXE; the sister disorder PEBEL1), NAXD constitutes the intracellular **NAD(P)HX repair system**, which converts the toxic, non-functional hydrated forms of the redox cofactors NADH and NADPH (collectively NAD(P)HX) back to usable NAD(P)H. When NAXD is deficient, S-NADHX, R-NADHX and cyclic-NADHX accumulate and functional NAD(P)H is depleted, producing a cellular energy/redox crisis ([PMID: 30576410](#); [PMID: 34161859](#)).

The defining clinical feature is a **gene–environment interaction**: the enzymopathy is often silent until a metabolic stressor — most commonly **fever or infection**, but also immunization, or physical trauma — sharply increases the non-enzymatic hydration of NAD(P)H and can denature thermolabile mutant enzyme. This precipitates acute, frequently fatal **neurometabolic decompensation**: rapidly progressive encephalopathy with brain and cerebellar edema and/or leukoencephalopathy, seizures, loss of developmental milestones, elevated CSF/serum lactate, characteristic **flexural necrotic skin lesions**, and — particularly for variants restricted to the mitochondrial isoform — **cardiomyopathy** and myopathy. Reported mortality across the combined NAXD/NAXE literature is approximately **78%**, with survivors experiencing neurological sequelae ([PMID: 39887790](#)).

Critically, the disorder is at least partly **treatable**: niacin/nicotinamide (vitamin B3), which feeds NAD *de novo* / salvage synthesis and replenishes the depleted cofactor pool, improves skin lesions and survival in reported patients, providing a rational metabolic bypass therapy alongside aggressive control of febrile triggers (PMID: 39887790; PMID: 27616477; PMID: 38974613). This report synthesizes 9 confirmed findings across 18 papers into a complete disease knowledge-base entry.

Evidence-source note. This is an ultra-rare disease first defined in 2019; knowledge is derived almost entirely from **aggregated case reports/series and functional studies** (human clinical, patient fibroblasts/iPSCs, HAP1 knockouts, zebrafish, recombinant-enzyme biochemistry), not from population EHR or registry data. Fewer than ~20 genetically confirmed NAXD patients have been published. Where a claim rests on the sister disorder NAXE (PEBEL1) or on model systems, this is stated explicitly.

1. Disease Information

Overview. NAXD deficiency is a rare metabolite-repair disorder. NAD(P)HX dehydratase is a highly conserved enzyme essential for intracellular repair of the damaged/hydrated redox cofactor NAD(P)HX. As stated in the founding case series: "*The highly conserved enzyme NAD(P)HX dehydratase (NAXD) is essential for intracellular repair of NAD(P)HX*" (PMID: 30576410). Loss of the enzyme produces a fever-triggered neurodegenerative and multisystem disease.

Key identifiers.

Resource	Identifier
Disease name	NAD(P)HX Dehydratase Deficiency / PEBEL2
MONDO	MONDO:0034121
OMIM (phenotype)	#618321 (Encephalopathy, progressive, early-onset, with brain edema and/or leukoencephalopathy, 2)
OMIM (gene)	615910
Gene symbol	NAXD (HGNC:25576)
NCBI Gene	55739
Ensembl	ENSG00000213995
UniProt	Q8IW45
Enzyme	EC 4.2.1.93 (ATP-dependent, ADP-forming NAD(P)HX dehydratase)
Locus	13q34 (GRCh38 chr13:110,615,505–110,643,086, + strand)
Aliases	CARKD, LP3298

The disease designation and OMIM ID are confirmed: *"Encephalopathy, progressive, early-onset, with brain edema and/or leukoencephalopathy, 2 (PEBEL2; MIM# 618321), caused by biallelic pathogenic variants in the NAD(P)HX dehydratase (NAXD) is a rare metabolite repair disorder"* (PMID: 34161859).

Synonyms / alternative names: PEBEL2; NAXD deficiency; NAD(P)HX dehydratase deficiency; carbohydrate kinase domain-containing protein deficiency (CARKD). The closely related sister disorder is **NAXE deficiency / PEBEL1** (OMIM #617186), caused by the partner repair enzyme.

Information source. Knowledge derives predominantly from **aggregated disease-level resources** (OMIM, Orphanet, MONDO) and from **individual patient case reports / small case series** in the primary literature, supplemented by in vitro (cell line, recombinant enzyme) and model-organism (zebrafish) data. There is no large EHR-derived cohort; this is an ultra-rare disorder documented through gene-first (WES/WGS) discovery.

2. Etiology

Primary cause — genetic. The disease is caused by **biallelic (homozygous or compound-heterozygous) loss-of-function variants in *NAXD***, inherited in an autosomal-recessive pattern. Biallelic *NAXD* variants were identified by whole-exome/whole-genome sequencing in a case series of infants and children with febrile-illness-induced neurodegeneration or cardiac failure and early death (PMID: 30576410). The molecular defect is a failure of an essential housekeeping "metabolite repair" function.

Genetic risk factors. The only established causal genetic factor is biallelic pathogenic *NAXD* variation. gnomAD constraint metrics (v2.1.1) are consistent with a recessive loss-of-function mechanism: $pLI \approx 5.5 \times 10^{-5}$ (i.e., ~ 0 , the gene tolerates heterozygous LoF) with observed/expected LoF (oe_lof/LOEUF) = 0.615 (95% CI 0.44–0.88). Carriers (heterozygotes) are asymptomatic.

Environmental "risk"/trigger factors. Uniquely, environment acts as the **decompensation trigger** rather than an independent cause. Documented triggers include **febrile illness and infection** (most common), **routine immunization** (PMID: 38214124), and **physical/mechanical stress such as mild head trauma** (PMID: 36834994). These raise body temperature and metabolic flux, accelerating non-enzymatic cofactor hydration and denaturing thermolabile mutant enzyme.

Protective factors. No genetic protective variants or modifier alleles have been defined. The main *modifiable* protective actions are **avoidance/aggressive management of fever and triggers** and **niacin/nicotinamide supplementation** (see Treatment).

Gene–environment interaction. This is a paradigmatic G×E disorder. The recombinant *NAXD* proteins bearing patient missense changes p.(Gly63Ser) and p.(Arg608Cys) were **thermolabile** with decreased V_{max} and increased K_M (PMID: 30576410) — providing a direct molecular explanation for why fever (a rise in body temperature) converts a compensated enzymopathy into an acute crisis. During stress, *"nonenzymatic conversion of NAD(P)H to NAD(P)HX increases, and in the absence of repair, NAD(P)H is depleted, and NAD(P)HX accumulates, leading to decompensation"* (PMID: 35637064).

3. Phenotypes

The phenotype spectrum spans neurological, cutaneous, cardiac, muscular and biochemical domains. Onset is typically in the first 1–3 years of life (median age of onset ~1.16 years across the combined cohort), though adult-onset is reported.

Phenotype	Type	Onset / severity / course	Frequency	Suggested HPO
Progressive encephalopathy / psychomotor regression	Clinical sign	Infancy–early childhood; severe; episodic-on-progressive, often fever-triggered	Very frequent	HP:0002376 (Developmental regression); HP:0006846 (Acute encephalopathy)
Brain / cerebellar edema	Physical manifestation (imaging)	Acute during crisis; severe	Frequent	HP:0002181 (Cerebral edema)
Leukoencephalopathy / white-matter changes	Imaging abnormality	Subacute–chronic; severe	Frequent	HP:0002352 (Leukoencephalopathy)
Seizures (incl. myoclonic)	Clinical sign	Infancy; variable	Frequent (whole-cell deficiency)	HP:0001250 (Seizure); HP:0002123 (Generalized myoclonic seizures)
Hypotonia (axial)	Clinical sign	Infancy; moderate–severe	Frequent	HP:0001252 (Hypotonia); HP:0008936 (Axial hypotonia)
Ataxia	Clinical sign	Childhood; variable	Frequent	HP:0001251 (Ataxia)
Flexural erythematous/ erosive/necrotic skin lesions	Physical manifestation	With crisis; severe	~31% of whole-cell deficiency	HP:0000988 (Skin rash); HP:0200041 (Skin erosion); HP:0100697 (Necrosis of the skin)
Cardiomyopathy / cardiac failure	Clinical sign	Infancy or later; severe	Subset (esp. mito-)	HP:0001638 (Cardiomyopathy);

Phenotype	Type	Onset / severity / course	Frequency	Suggested HPO
			isoform variants)	HP:0001635 (Congestive heart failure)
Myopathy / neuropathy	Clinical sign	Variable; moderate	Mito-isoform variants	HP:0003198 (Myopathy); HP:0009830 (Peripheral neuropathy)
Elevated CSF/serum lactate	Laboratory abnormality	During/after crisis; mild-marked	Frequent	HP:0002151 (Increased serum lactate); HP:0002490 (Increased CSF lactate)
Respiratory insufficiency	Clinical sign	Crisis; severe	Subset	HP:0002093 (Respiratory insufficiency)
Ophthalmoparesis	Clinical sign	Variable	Subset	HP:0000602 (Ophthalmoplegia)
Premature death	Outcome	Early childhood typical	~78% mortality	HP:0001522 (Death in infancy)

Characteristic cutaneous phenotype: *"The characteristic skin eruption comprises well-demarcated erythematous and erosive plaques progressing to blistering and necrosis, predominantly affecting flexural surfaces"* (PMID: 39887790).

Phenotypic diversity / atypical presentations. Not all patients show the full picture. One 7-month-old with a novel homozygous variant had neither preceding fever nor skin lesions, prompting the authors to note that *"cases show phenotypic diversity"* (PMID: 36158054). A patient presenting after routine immunizations had prominent skin findings **in the absence of fevers** (PMID: 38214124).

Quality-of-life impact. Formal QoL instruments (EQ-5D/SF-36) have not been applied to this ultra-rare disease. Qualitatively, the impact is profound: acute crises cause loss of acquired milestones, severe disability in survivors, and high early mortality.

4. Genetic / Molecular Information

Causal gene. *NAXD* (HGNC:25576; NCBI Gene 55739; Ensembl ENSG00000213995; gene-OMIM 615910), located at **13q34**, encoding NAD(P)HX dehydratase (UniProt Q8IW45; EC 4.2.1.93). The sister gene is *NAXE* (HGNC:18453; Gene 128240; ENSG00000163382; 1q22; UniProt Q8NCW5; EC 5.1.99.6; phenotype-OMIM 617186; aliases APOA1BP/AIBP/YJEFN1).

Pathogenic variant spectrum. Variants are biallelic and predominantly **missense**, with frameshift/loss-of-function alleles also reported. Representative documented variants:

Variant (cDNA / protein)	Zygosity	Functional evidence	Reference
p.(Gly63Ser) and p.(Arg608Cys)	Compound het (recombinant)	Thermolabile; ↓ Vmax, ↑ KM for ATP-dependent NADHX dehydratase activity	PMID: 30576410
c.301G>A, p.(Ala101Thr)	Homozygous	Novel missense via exome sequencing	PMID: 34161859
c.247G>A	Homozygous	Novel; myoclonic seizures, no fever/skin lesions	PMID: 36158054
c.101_102delTA, p.(Thr35Phefs*63) + c.318C>G, p.(Ile160Met)	Compound het	Used to derive patient iPSC line	PMID: 38387170
c.362C>T, p.(Pro121Leu)	VUS	Probable PEBEL2, post-immunization skin findings	PMID: 38214124
Compound-heterozygous (cardiomyopathy)	Compound het	Metabolic cardiomyopathy with interstitial fibrosis	PMID: 39822994

The founding study demonstrated loss of function at the protein level: *"Recombinant NAXD protein harbouring two missense variants leading to the amino acid changes p.(Gly63Ser) and p.(Arg608Cys) were thermolabile and showed a decrease in Vmax and increase in KM for the ATP-dependent NADHX dehydratase activity"* (PMID: 30576410).

Variant classification & population frequency. Reported variants are classified pathogenic/likely-pathogenic per ACMG (with occasional VUS such as p.Pro121Leu). Pathogenic alleles are exceedingly rare in gnomAD, consistent with a severe recessive disorder; heterozygous LoF is tolerated (pLI≈0), while biallelic LoF is disease-causing.

Functional consequence — loss of function (reduced/abolished enzymatic repair activity), not gain of function or dominant negative.

Isoform-determined genotype–phenotype correlation (modifier of expression). *NAXD* encodes two subcellular isoforms: *"Exon 1 of NAXD contains a mitochondrial propeptide, and a unique cytosolic isoform is initiated from an alternative start codon in exon 2"* (PMID: 35866541). The isoform hit by a given variant dictates phenotype: variants affecting **both** isoforms → neurological degeneration, seizures and skin lesions; variants affecting **only the mitochondrial** isoform → *"myopathy, moderate neuropathy and a cardiac presentation, without the characteristic skin lesions, seizures or neurological degeneration"* (PMID: 35866541).

Modifier genes / epigenetics / chromosomal abnormalities. No trans-acting modifier genes, epigenetic mechanisms, or large-scale chromosomal abnormalities have been implicated; the "modifier" of expression is the intragenic isoform architecture described above.

5. Environmental Information

- **Environmental triggers (not toxins per se):** febrile illness, infection, immunization, and physical/mechanical stress (e.g., mild head trauma). These accelerate the endogenous chemical damage reaction ($\text{NAD(P)H} \rightarrow \text{NAD(P)HX}$). NAD(P)HX itself is described as *"a toxic metabolite that is produced by stressors such as a fever, infection, or physical stress"* (PMID: 38214124).
- **Lifestyle factors:** none established; disease presents in infancy/childhood. Relevant management is trigger avoidance and prompt antipyresis.
- **Infectious agents:** no specific pathogen causes the disease; **any** febrile infection can act as a non-specific trigger of decompensation.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function variants in *NAXD* → loss of ATP-dependent NAD(P)HX dehydratase activity** (demonstrated: thermolabile recombinant mutants with reduced V_{max} /increased K_M ; PMID: 30576410).

2. Loss of repair capacity **leads to** failure to convert the damaged, hydrated cofactors back to NAD(P)H → **accumulation of S-NADHX, R-NADHX and cyclic-NADHX** and **depletion of functional NAD(P)H** (demonstrated in patient fibroblasts: *"highly elevated concentrations of the damaged cofactors S-NADHX, R-NADHX and cyclic NADHX"*; [PMID: 30576410](#)).
3. A **metabolic stressor (fever/infection/immunization/trauma)** results in an increased rate of non-enzymatic NAD(P)H hydration **and** denaturation of thermolabile mutant enzyme → an unrepaired surge in NAD(P)HX and acute cofactor depletion (*"During stress, nonenzymatic conversion of NAD(P)H to NAD(P)HX increases...leading to decompensation"*; [PMID: 35637064](#)).
4. This **branches** into downstream lesions:
5. **4a.** Accumulated NAD(P)HX **inhibits multiple dehydrogenases** and NAD(P)H is depleted → impaired redox/energy metabolism → **mitochondrial dysfunction** (higher sensitivity to metabolic stress in galactose/azide media; [PMID: 30576410](#)).
6. **4b.** Repair failure **results in strong inhibition of the cytosolic, de novo serine synthesis pathway** — a distinct, recently discovered downstream mechanism beyond mitochondrial impairment (*"metabolomic analyses revealed a strong inhibition of the cytosolic, de novo serine synthesis pathway"*; [PMID: 39789421](#)).
7. Combined energetic/redox and biosynthetic failure **injures high-demand tissues** → **brain/cerebellum/white matter** (encephalopathy, edema, leukoencephalopathy, seizures), **heart/muscle** (cardiomyopathy, myopathy), and **skin** (flexural erosive/necrotic lesions), with **elevated CSF/serum lactate** as a biochemical readout.
8. Untreated, the crisis **progresses to** coma, brain atrophy and **death** (~78% mortality); metabolic bypass with niacin/nicotinamide **partially reverses** the cofactor deficit and improves skin and survival.

Detail by category

- **Molecular pathway / biochemical abnormality:** defect in the **NAD(P)HX repair (metabolite-repair) pathway**; NAXD (EC 4.2.1.93) works with NAXE (EC 5.1.99.6). The hydrated cofactors are inhibitors of several dehydrogenases and generate harmful byproducts.
- **Metabolic changes:** depletion of the NAD(P)H redox pool; impaired energy metabolism; **inhibition of de novo serine synthesis** ([PMID: 39789421](#)); elevated lactate reflecting a shift toward anaerobic metabolism / mitochondrial impairment.

- **Cellular processes:** mitochondrial dysfunction; vulnerability under oxidative/metabolic stress (demonstrated by growth impairment in galactose vs glucose; [PMID: 39789421](#), [PMID: 30576410](#)); neurodegeneration.
- **Protein dysfunction:** thermolability of mutant enzyme (mechanistic link to fever); reduced catalytic efficiency ($\downarrow V_{max}$, $\uparrow K_M$).
- **Immune involvement:** zebrafish *naxd* knockouts reveal **immune-system perturbations in early development** ([PMID: 41621837](#)); clinically, infection/immunization acts as trigger, and inflammation likely amplifies cofactor-hydration stress.
- **Tissue-damage mechanisms:** metabolic/redox stress leading to cytotoxic edema, necrosis (skin), fibrosis (cardiac interstitial fibrosis; [PMID: 39822994](#)).

Suggested GO / CHEBI / CL terms: GO:0110051 (metabolite repair) / GO:0046496 (nicotinamide nucleotide metabolic process); GO:0052855 (ADP-dependent NAD(P)H-hydrate dehydratase activity, EC 4.2.1.93); GO:0006564 (L-serine biosynthetic process); GO:0005739 (mitochondrion, CC); GO:0005829 (cytosol, CC). CHEBI: NADHX, NADPHX, NAD(H), NADP(H), nicotinamide (CHEBI:17154), niacin (CHEBI:15940). CL: neuron (CL:0000540), cardiomyocyte (CL:0000746), keratinocyte (CL:0000312).

7. Anatomical Structures Affected

- **Organ level (primary):** brain and cerebellum (UBERON:0000955 brain; UBERON:0002037 cerebellum), cerebral white matter (UBERON:0002316), spinal cord (UBERON:0002240; myelopathy reported in the NAXE sister disorder). **Skin** (UBERON:0002097), predominantly flexural surfaces. **Heart** (UBERON:0000948) and skeletal muscle (UBERON:0001134) in cardiac/myopathic presentations.
- **Body systems:** central and peripheral **nervous system**, **integumentary system**, **cardiovascular system**, **musculoskeletal system**; **respiratory** insufficiency during crisis.
- **Tissue/cell level:** neurons and glia (white-matter/leukoencephalopathy → oligodendrocyte-supported myelin); cardiomyocytes (with interstitial fibrosis); keratinocytes/epidermis (necrotic skin lesions).
- **Subcellular level:** **mitochondrion** (GO:0005739) and **cytosol** (GO:0005829) — the two compartments corresponding to the mitochondrial and cytosolic NAXD isoforms; this compartmentalization directly maps to phenotype (Section 4).

- **Lateralization:** central lesions are typically **bilateral/symmetric** (edema, leukoencephalopathy); skin lesions are bilateral and flexural.

8. Temporal Development

- **Onset:** typically **congenital-to-early-childhood**, most within the first 1–3 years; median age of onset ~1.16 years in the combined NAXD/NAXE cohort ([PMID: 39887790](#)). **Adult onset is possible:** a 32-year-old presented after mild head trauma ([PMID: 36834994](#)).
- **Onset pattern:** characteristically **(sub)acute** decompensation superimposed on a previously well or mildly affected child, precipitated by a trigger.
- **Progression: episodic-on-progressive** — stepwise deterioration with each febrile/stress event, often **rapidly progressive** to coma and death without intervention. Between crises, patients may be relatively stable.
- **Disease course / duration:** frequently **fatal in early childhood**; survivors have chronic, lifelong neurological sequelae.
- **Remission:** no spontaneous cure; **treatment-induced stabilization/improvement** is reported with niacin/nicotinamide and trigger control.
- **Critical periods / window of opportunity:** the peri-febrile window is both the period of maximal vulnerability and the key intervention window — prompt antipyresis, supportive care and NAD-precursor supplementation.

9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic *NAXD* variants). Both parents are asymptomatic carriers.
- **Penetrance/expressivity:** appears **high penetrance in the biallelic state** but **variable expressivity** — modulated by which isoform is affected (Section 4) and by exposure to triggers; some homozygotes present atypically without fever or skin lesions ([PMID: 36158054](#)).
- **Epidemiology: ultra-rare;** only case reports and small series (tens of patients worldwide across NAXD and NAXE). No reliable prevalence/incidence estimate exists; Orphanet lists it

among ultra-rare metabolic disorders. A literature review compiled **45 NAXD/NAXE patients with 31 pathogenic/likely-pathogenic mutations** ([PMID: 39887790](#)).

- **Founder effects / consanguinity:** homozygous variants in reported patients suggest a role for **consanguinity** in some families; no defined founder allele has been reported.
- **Carrier frequency:** not formally established; consistent with rarity, gnomAD shows very low pathogenic-allele frequencies.
- **Population demographics / geography:** cases reported across diverse populations (Europe, Middle East, Asia including China; a Chinese cardiomyopathy case is noted as "one of the few" in China; [PMID: 39822994](#)). No strong sex bias is established (both sexes affected). Age distribution skews to infancy/early childhood.

10. Diagnostics

Laboratory tests / biomarkers. - **Elevated CSF and/or serum lactate** is a key, reproducible biochemical clue (mild-to-marked; present in the NAXE sister disorder in all affected individuals: *"Lactate was elevated in cerebrospinal fluid of all affected individuals"*; [PMID: 27616477](#)). Basal metabolic tests may otherwise be near-normal ([PMID: 36158054](#)). - **Definitive biomarker:** markedly elevated damaged cofactors **S-NADHX, R-NADHX and cyclic-NADHX** in patient fibroblasts (research/specialized assay), abrogated by wild-type *NAXD* rescue ([PMID: 30576410](#)). - **LOINC:** lactate CSF (LOINC 2519-7), lactate plasma/serum (LOINC 2524-7 / 32693-4).

Imaging. Brain **MRI** showing cerebral/cerebellar **edema, leukoencephalopathy**, and (in progression) global brain atrophy is central to recognizing the acute encephalopathy.

Cardiac work-up. Echocardiography/cardiac MRI and endomyocardial evaluation may reveal metabolic **cardiomyopathy with interstitial fibrosis** in the absence of coronary disease or hypertension ([PMID: 39822994](#)).

Genetic testing (diagnostic gold standard). Diagnosis is molecular. **Whole-exome or whole-genome sequencing** identifies biallelic *NAXD* variants (the discovery and most subsequent diagnoses used WES/WGS; [PMID: 30576410](#); [PMID: 34161859](#)). Targeted single-gene/panel testing (mitochondrial/leukodystrophy/metabolic-encephalopathy panels including *NAXD* and *NAXE*) is appropriate when the phenotype is suggestive. Segregation confirms biallelic status. Functional confirmation (fibroblast NAD(P)HX measurement or recombinant enzyme assay) can resolve VUS.

Clinical criteria / differential diagnosis. No formal consensus criteria exist; diagnosis rests on the triad of **fever-triggered neuroregression + suggestive MRI (edema/leukoencephalopathy) + elevated lactate**, confirmed genetically. Because it can be mistaken for a **primary mitochondrial disease** (patients have been treated with a "mitochondrial cocktail"; [PMID: 36158054](#)), key differentials include Leigh syndrome and other mitochondrial encephalopathies, other leukodystrophies, biotin-thiamine-responsive basal ganglia disease, and — with skin involvement — nutritional/genetic niacin-deficiency states (pellagra-like). The sister disorder **NAXE deficiency (PEBEL1)** is the closest differential and is distinguished by gene.

Screening. Not part of routine newborn screening. **Cascade carrier testing** of relatives and **prenatal/preimplantation genetic testing** are available once the familial variants are known.

11. Outcome / Prognosis

- **Mortality:** high — *"The mortality rate was 78%, with survivors experiencing varying degrees of neurological sequelae"* ([PMID: 39887790](#)). In the NAXE sister disorder, the course was rapidly progressive to coma, global brain atrophy and death.
 - **Morbidity/disability:** survivors typically have significant neurological disability (developmental regression, spasticity, epilepsy, motor impairment).
 - **Prognostic factors: which isoform is affected** (mitochondrial-only vs whole-cell) shapes organ involvement and course; **frequency/severity of triggering events**; and **early institution of niacin/nicotinamide and trigger control**, which improve skin lesions and survival. Niacin response, however, is **not universal**: one PEBEL1 patient deteriorated to a fatal outcome despite the highest reported niacin dose ([PMID: 38974613](#)).
 - **Complications:** status epilepticus, respiratory failure, cardiac failure, secondary infections, and sequelae of brain injury.
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12. Treatment

There is no curative therapy; management is **metabolic bypass + aggressive trigger control + supportive care**.

Pharmacotherapy / metabolic bypass. - **Niacin / nicotinamide (vitamin B3)** — the principal disease-modifying agent. By feeding NAD *de novo*/salvage synthesis it replenishes the depleted NAD(P) pool: "*Niacin/nicotinamide supplementation resulted in improvements in skin lesions and survival rates*" (PMID: 39887790). The rationale is explicit in the NAXE literature: "*NAD or nicotinic acid (vitamin B3) supplementation might have therapeutic implications for this fatal disorder*" (PMID: 27616477). A systematic review of 7 PEBEL1/PEBEL2 patients found most improved or stabilized on niacin, though one deteriorated fatally (PMID: 38974613). NCIT: niacin (NCIT:C574), nicotinamide (NCIT:C577). - **Adverse-event management:** niacin-related urticaria has been managed off-label with a COX-2 inhibitor (PMID: 38974613).

Acute supportive care. Prompt **antipyresis** and **treatment of the precipitating infection**, intensive supportive/neurocritical care during crises, seizure management, and cardiac support as needed. Empirical **mitochondrial "cocktail"** has been used but is not specifically corrective (PMID: 36158054).

Advanced / experimental therapeutics. No approved gene, cell, or RNA therapy exists. Wild-type *NAXD* lentiviral rescue corrects the biochemical defect in patient fibroblasts (proof of concept for gene replacement; PMID: 30576410). NAD-precursor strategies (nicotinamide riboside, nicotinic acid) are of mechanistic interest.

Personalized approach. Genotype (isoform affected) and trigger history should guide monitoring (e.g., cardiac surveillance for mitochondrial-isoform variants) and preventive planning.

13. Prevention

- **Primary prevention:** for at-risk families, **genetic counseling**, carrier testing, and reproductive options (prenatal/PGT) prevent recurrence. There is no population-level primary prevention (not vaccine-preventable; disease is genetic).
- **Secondary prevention (trigger mitigation):** in known-affected individuals, **prompt and aggressive control of fever/infection**, vigilance around immunizations and physical stress, and **prophylactic niacin/nicotinamide** aim to prevent decompensation. Caution and close monitoring are warranted around immunization given a reported post-vaccination decompensation (PMID: 38214124).
- **Tertiary prevention:** rehabilitation, epilepsy management, cardiac and respiratory support to limit complications in survivors.

- **Counseling:** autosomal-recessive recurrence risk is **25%** per pregnancy for carrier couples; cascade testing of relatives is recommended.
 - **Screening:** cascade carrier screening; prenatal diagnosis where familial variants are known. Not currently in newborn-screening panels.
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14. Other Species / Natural Disease

- **Evolutionary conservation:** the NAD(P)HX repair system is **ancient and highly conserved**. NAXD/NAXE orthologs function in plants (*Arabidopsis thaliana*, *Zea mays*), *E. coli*, and vertebrates: "*Arabidopsis thaliana* and *Zea mays* NAD(P)HX dehydratase (NAXD) and NAD(P)HX epimerase (NAXE), two enzymes that are involved in repair of chemically damaged NAD(P)H cofactors" ([PMID: 36710015](#)).
 - **Orthologs / taxonomy (NCBI Taxon):** human *NAXD* (Gene 55739); zebrafish *naxd* (used for CRISPR models; *Danio rerio*, taxon 7955); plant orthologs in *A. thaliana* (taxon 3702) and *Z. mays* (taxon 4577); bacterial *yjeF*-family in *E. coli*.
 - **Natural disease in other species:** no spontaneously occurring NAXD-deficiency disease has been catalogued in companion animals/wildlife (no OMIA entry noted); relevance to date is via engineered models.
 - **Zoonotic potential:** not applicable (genetic, non-transmissible).
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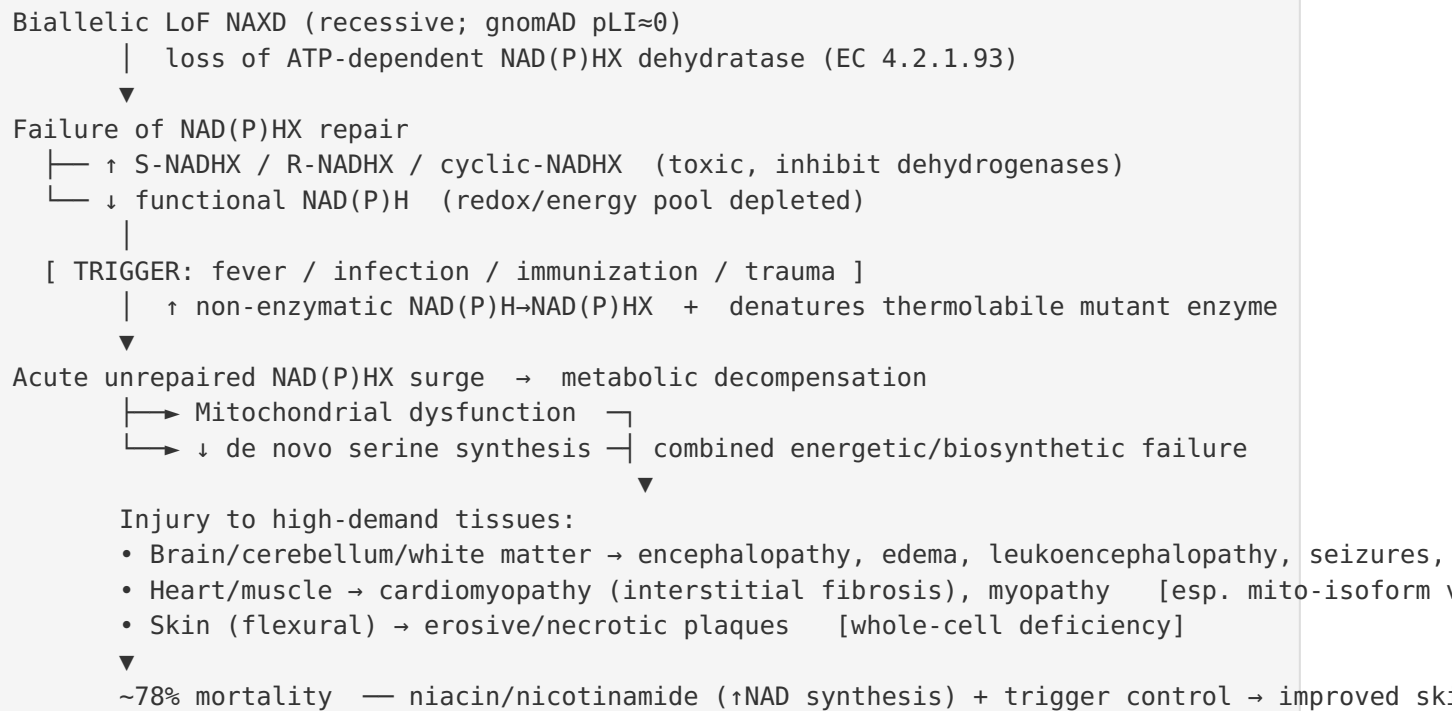
15. Model Organisms

Model	Type	Key features / recapitulation	Reference
Zebrafish <i>naxd</i> (and <i>naxe</i>) CRISPR/Cas9 knockouts	Vertebrate, in vivo	Both accumulate NADHX; <i>naxd</i> line shows distinctive features and immune-system perturbations in early development	PMID: 41621837
Human HAP1 NAXD knockout	Cellular, in vitro	Growth impairment specifically in galactose vs glucose; metabolomics reveals de novo serine synthesis inhibition ; models the metabolic lesion	PMID: 39789421
Patient fibroblasts	Primary human cells	Elevated S-/R-/cyclic-NADHX; corrected by WT NAXD rescue; mitochondrial-stress sensitivity	PMID: 30576410
Patient iPSC line BCHNDi001-A	iPSC	Derived from PEBEL2 fibroblasts (c.101_102delTA; c.318C>G); enables differentiation into affected lineages	PMID: 38387170
Recombinant NAXD/NAXE (E. coli; plant systems)	In vitro enzymology	Purified enzymes for kinetic/thermostability assays (demonstrated thermolability of mutants)	PMID: 30576410; PMID: 36710015

Zebrafish CRISPR evidence: "we generated zebrafish lines deficient in *naxe* or *naxd* using CRISPR/Cas9 technology. While both models accumulated NADHX, only *naxd*..." (PMID: 41621837). **iPSC evidence:** "we generated an induced pluripotent stem cell (iPSC) line from the dermal fibroblasts (HDFs) of a PEBEL2 patient who carried biallelic mutations, c.101_102delTA(p.Thr35Phefs63) and c.318C > G (p.Ile160Met) in NAXD"* (PMID: 38387170).

Model applications & limitations: these systems recapitulate the core **NADHX accumulation** and metabolic lesion and enable therapeutic testing; limitations include incomplete recapitulation of the full human multisystem/fever-triggered clinical phenotype and species-specific differences. Resources: ZFIN (zebrafish), Cellosaurus (HAP1, iPSC lines).

Mechanistic Model / Interpretation



Upstream vs downstream: the *NAXD* mutation and cofactor-repair failure are **upstream**; mitochondrial dysfunction and serine-synthesis inhibition are **parallel downstream** effectors; tissue injury and clinical crisis are **terminal**. The **fever/thermolability node** is the key modifiable amplifier converting a compensated enzymopathy into acute disease, and the **niacin bypass** is the key therapeutic lever.

Evidence Base

PMID	Title (abbrev.)	Contribution
30576410	<i>NAXD deficiency: a novel neurodegenerative disorder exacerbated by febrile illnesses</i>	Founding disease description; recessive WES/WGS diagnosis; elevated S-/R-/cyclic-NADHX; thermolabile mutant enzyme ($\downarrow V_{max}$, $\uparrow K_M$); mitochondrial-stress sensitivity; WT rescue
34161859	<i>NAXD deficiency due to a novel biallelic missense variant + review</i>	Disease name/OMIM #618321; recessive basis; novel c.301G>A p.(Ala101Thr)
35866541	<i>Clinical/biochemical distinctions for NAXD or NAXE deficiency</i>	Isoform architecture (mito propeptide exon 1 vs cytosolic exon 2) $\rightarrow$ genotype–phenotype correlation
39789421	<i>Failure to repair NAD(P)H blocks de novo serine synthesis</i>	HAP1 KO galactose sensitivity; serine-synthesis inhibition mechanism
39887790	<i>Cutaneous manifestations of NAXD/NAXE deficiency</i>	45-patient review; ~78% mortality; flexural necrotic skin phenotype; niacin improves skin/survival
35637064	<i>NAXE deficiency amenable for metabolic correction</i>	Stress-triggered decompensation mechanism (NAD(P)H depletion)
27616477	<i>NAXE mutations cause a lethal neurometabolic disorder</i>	Sister disorder (PEBEL1): febrile-triggered ataxia/edema/skin; elevated CSF lactate; niacin rationale
41621837	<i>Zebrafish models of NADHX repair deficiency</i>	naxd/naxe CRISPR models; NADHX accumulation; immune perturbation
38387170	<i>iPSC line BCHNDi001-A from PEBEL2 patient</i>	Patient-derived iPSC with defined biallelic variants
39822994	<i>Metabolic cardiomyopathy from compound-het NAXD</i>	Cardiac phenotype with interstitial fibrosis
36834994	<i>Severe NAXD syndrome in adulthood after mild head trauma</i>	Adult onset; non-febrile physical-stress trigger
36158054	<i>A case with NAXD deficiency (novel c.247G>A)</i>	Phenotypic diversity; atypical case without fever/skin lesions; mild lactate elevation
38214124		

PMID	Title (abbrev.)	Contribution
	<i>Progressive encephalopathy after 4-month immunizations</i>	Immunization as trigger; skin findings without fever; VUS c.362C>T
38974613	<i>Transient response to high-dose niacin (NAXE)</i>	Niacin efficacy but not universal (one fatal outcome despite highest dose)
36710015	<i>Systems for plant protein expression</i>	Evolutionary conservation of NAXD/NAXE (Arabidopsis, maize)

Evidence source types: human clinical (case reports/series, literature reviews), in vitro (patient fibroblasts, HAP1 KO, recombinant enzyme, iPSC), model organism (zebrafish), and computational/genomic constraint (gnomAD).

Limitations and Knowledge Gaps

- **Ultra-rare, no cohort epidemiology.** Prevalence/incidence, carrier frequency, sex ratio, and penetrance estimates are unavailable; all clinical data come from small series prone to ascertainment/publication bias toward severe cases (mortality may be overestimated).
- **Treatment evidence is anecdotal.** Niacin/nicotinamide benefit is based on case reports and one small systematic review; there are **no randomized trials**, optimal dosing is undefined, and at least one patient deteriorated despite maximal therapy ([PMID: 38974613](#)).
- **Mechanistic gaps.** The relative contributions of mitochondrial dysfunction versus serine-synthesis inhibition to specific organ phenotypes are not resolved; why skin and cerebellum are especially vulnerable is unexplained; the immune-perturbation finding is model-derived and needs human validation.
- **Genotype–phenotype nuance.** The isoform rule explains broad patterns but individual variability (atypical fever-negative cases) indicates unidentified modifiers.
- **No natural animal disease / limited long-term outcome data.**

Proposed Follow-up Experiments / Actions

1. **Establish an international patient registry** to define natural history, prevalence, penetrance, sex ratio, and standardized outcomes; harmonize trigger and treatment data.
 2. **Prospective, dose-finding evaluation of NAD precursors** (niacin, nicotinamide, nicotinamide riboside) with biochemical endpoints (NADHX/NAD(P)H, lactate) — ideally a basket trial spanning PEBEL1/PEBEL2.
 3. **iPSC-derived organoid modeling** (neuronal, cardiac, keratinocyte) from BCHNDi001-A and new lines to test whether serine supplementation and NAD precursors rescue tissue-specific phenotypes ([PMID: 38387170](#); [PMID: 39789421](#)).
 4. **Thermal-stress assays across the variant panel** to build a genotype→thermolability→clinical-severity map, informing prognosis and trigger-avoidance counseling ([PMID: 30576410](#)).
 5. **Gene-replacement proof-of-concept in vivo** (zebrafish *naxd* and/or AAV-NAXD in mammalian models), building on fibroblast lentiviral rescue.
 6. **Dissect the immune-perturbation phenotype** from zebrafish in human systems to test whether inflammation independently amplifies decompensation ([PMID: 41621837](#)).
 7. **Develop a clinical NAD(P)HX/serine metabolite biomarker assay** for rapid diagnosis and treatment monitoring, and evaluate inclusion of *NAXD/NAXE* in mitochondrial/leukodystrophy gene panels and consideration for newborn screening pilots.
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Report compiled from 9 confirmed findings across 18 reviewed papers. All quoted material is verbatim from the cited PubMed abstracts.
